

# Master Enterprise Architecture Specification

Complete Technical Overview of the RVCC Digital Procurement & ERP Ecosystem

MASTER ARCHITECTURE

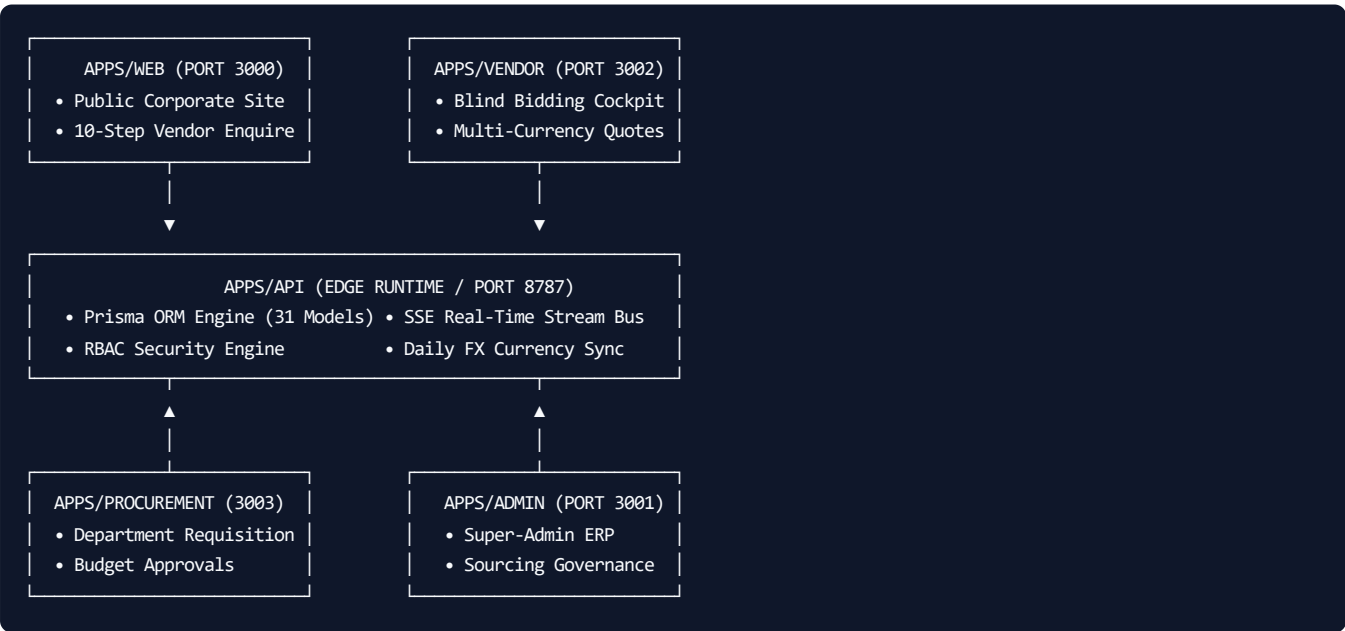
|                                          |                                    |                                                 |                                  |
|------------------------------------------|------------------------------------|-------------------------------------------------|----------------------------------|
| <b>Project:</b> RVCC Enterprise Monorepo | <b>Version:</b> 2.0.0 (Production) | <b>Architecture:</b> Next.js 16 + Edge + Prisma | <b>Status:</b> Verified & Active |
|------------------------------------------|------------------------------------|-------------------------------------------------|----------------------------------|

## 1. Master System Architecture & Enterprise Topology

The **RVCC Enterprise Digital Ecosystem** represents a state-of-the-art multi-tier procurement and digital transformation platform. Built from the ground up to replace fragmented legacy workflows, it unifies corporate marketing, supplier qualification, real-time blind bidding auctions, internal purchase requisitions, and administrative oversight into a cohesive, high-performance monorepo architecture.

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                              |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Architectural Highlights</b> <ul style="list-style-type: none"><li>• <b>5 Specialized Applications:</b> Web, Admin, Vendor, Procurement, and Edge API.</li><li>• <b>3 Shared Internal Packages:</b> Centralized Zod schemas, TypeScript types, and utilities.</li><li>• <b>Prisma ORM &amp; PostgreSQL:</b> 31 relational models with soft-delete safeguards.</li><li>• <b>Edge-Ready Infrastructure:</b> Cloudflare Workers + Vercel multi-cloud deployment.</li></ul> | <b>Production Quality Metrics</b> <ul style="list-style-type: none"><li>• <b>Lines of Code:</b> 65,453 lines of clean, strictly typed TypeScript.</li><li>• <b>Codebase Files:</b> 580 production files across 8 monorepo packages.</li><li>• <b>Testing &amp; CI:</b> 10/10 automated Vitest suites passing in CI pipelines.</li><li>• <b>Code Formatting:</b> 100% Prettier + Tailwind class sorting compliance.</li></ul> |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## 2. End-to-End Enterprise Data Flow



### 3. Summary Index of Technical Specifications

| Document Code | Specification Title                                                  | Target Application / Scope                                                        |
|---------------|----------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| SPEC-01       | <a href="#">Monorepo &amp; Shared Architecture Specification</a>     | Turborepo, PNPM Workspaces, Shared Packages, CI/CD, Tooling & PWA.                |
| SPEC-02       | <a href="#">Public Web &amp; Vendor Onboarding Specification</a>     | <code>apps/web</code> : 3D WebGL, 10-Step Supplier Wizard, Careers, Flipbook.     |
| SPEC-03       | <a href="#">Admin ERP &amp; Sourcing Governance Specification</a>    | <code>apps/admin</code> : RBAC, RFQ Sourcing, Live Monitor, Supplier Review, CMS. |
| SPEC-04       | <a href="#">Vendor Portal &amp; Blind-Bidding Specification</a>      | <code>apps/vendor</code> : Blind Bidding Cockpit, SSE Stream, Multi-Currency FX.  |
| SPEC-05       | <a href="#">Procurement &amp; Purchase Requisition Specification</a> | <code>apps/procurement</code> : Department Requisitions, Budget Approval Matrix.  |
| SPEC-06       | <a href="#">API &amp; Database Architecture Specification</a>        | <code>apps/api</code> : 31 Prisma Models, PostgreSQL, Edge Runtime, SSE Bus.      |

### 4. Security, Compliance & Governance Standards

- **Zero-Trust Authentication:** Strict HttpOnly, Secure, SameSite=Lax JWT session cookies preventing XSS token theft.
- **Cryptographic Password & OTP Security:** Timing-safe comparison and SHA-256 token hashing for all authentication and password-reset workflows.
- **Data Integrity & Auditability:** Immutable `AuditLog` records capturing user identity, IP address, timestamp, action type, and before/after state diffs for all financial, sourcing, and administrative operations.
- **Soft Deletion:** Logical deletion across all primary business entities preventing accidental or malicious data loss.